

MASK WHAT MATTERS: CONTROLLABLE TEXT-GUIDED MASKING FOR SELF-SUPERVISED MEDICAL IMAGE ANALYSIS

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Anonymous submission

ABSTRACT

Annotated data scarcity remains a central obstacle to training reliable vision models in specialized domains such as medical imaging. Self-supervised masked image modeling (MIM) directly addresses this bottleneck, but dominant methods still depend on random, high-ratio masking, which wastes capacity on redundant image content and weakly aligns pretraining with clinically meaningful semantics. Existing region-aware variants only partially resolve this gap because they often rely on reconstruction heuristics or supervised signals that constrain transfer across tasks and modalities. We introduce Mask What Matters, a controllable text-guided masking framework for self-supervised medical image analysis. The framework uses vision-language models to localize prompt-specified regions and then applies differentiated masking that deliberately emphasizes diagnostically relevant content while suppressing redundant background reconstruction. This design makes semantic control an explicit component of MIM and yields better aligned representations with stronger cross-task generalization. Across brain MRI, chest CT, and lung X-ray, Mask What Matters consistently improves over established MIM methods such as SparK, with gains of up to +3.1 percentage points in classification accuracy, +1.3 in box average precision (BoxAP), and +1.1 in mask average precision (MaskAP) for detection. Crucially, these gains are obtained with substantially lower overall masking ratios (e.g., 40% vs. 70%), showing that semantically targeted masking is more effective than simply masking more patches. The results establish controllable, text-driven masking as a decisive mechanism for semantically aligned self-supervised learning and a practical route to stronger medical image representations.

1 INTRODUCTION

Image classification, segmentation, and object detection form the foundation of modern computer vision. Deep learning now drives the strongest progress on these tasks, yet its best-performing models still depend heavily on large annotated datasets. In specialized domains such as medical image processing, where expert annotations are scarce, expensive, and time-consuming to obtain, this dependence becomes a primary barrier to deploying robust visual models at scale.

Self-supervised learning (SSL) directly reduces this annotation burden by pretraining models on unlabeled data and learning transferable representations without manual labels [Jaiswal et al. (2020); Liu et al. (2021); Krishnan et al. (2022); Huang et al. (2023)]. Within SSL, MIM has become a particularly influential paradigm. By masking image content and requiring reconstruction from visible context, MIM forces models to capture structural and semantic regularities that transfer to downstream tasks. Its success across general vision has also translated to medical image analysis [He et al. (2022); Gupta et al. (2025); Xiao et al. (2023)].

Despite this progress, prevailing MIM approaches, including masked autoencoders (MAE) [He et al. (2022)] and SparK [Tian et al. (2023)], typically rely on random masking that is not aligned with the regions downstream tasks actually use. This task-agnostic strategy creates two concrete limitations: (1) learned representations can be semantically misaligned with downstream objectives, weakening transfer performance. (2) To increase the probability of masking informative regions, existing meth-

ods often adopt very high masking ratios (e.g., 75% in MAE), which makes reconstruction harder and raises data and computation demands. By contrast, natural language has compact structure and high semantic density, enabling effective learning even at low masking ratios [Devlin et al., (2019)]. The persistent need for high masking ratios in image-based MIM reflects visual redundancy and, more importantly, the absence of semantic-aware masking that directs the model toward genuinely critical regions.

Medical imaging makes this weakness especially consequential. Diagnostic evidence is often semantically sparse: lesions, organs, or pathological structures occupy limited regions, while large background areas contribute little to the target decision. At the same time, medical images span MRI, CT, X-ray, and other modalities, with task-relevant regions varying substantially in shape, size, and location. Effective medical MIM therefore requires masking strategies that are not only efficient, but also controllable and semantically generalizable.

We address these challenges with a controllable, text-guided masking framework, **Mask-What-Matters (MWM)**. Given a user-specified open-vocabulary phrase or description, a pretrained vision-language model supplies zero-shot localization cues that MWM converts into robust regions of interest (ROIs). MWM then performs prompt-conditioned, region-specific masking, assigning higher masking ratios to semantically important areas such as lesions or organs and lower ratios to background. In doing so, MWM injects downstream task semantics directly into the pretraining objective without requiring per-image reports or labels.

Our contributions are direct and concrete:

- We present the first controllable text-guided masking framework for medical imaging. MWM integrates vision-language models (BiomedCLIP) with segmentation refinement (SAM) to localize task-relevant regions from open-vocabulary prompts and apply differentiated ROI vs. background masking, directly correcting the semantic misalignment of random masking.
- We deliver an annotation-free and backbone-agnostic design. MWM requires no per-image reports or labels, and it integrates cleanly with both ViT- and ConvNet-based MIM pipelines under a unified protocol.
- We demonstrate consistent gains across modalities and tasks. On brain MRI, chest CT, and X-ray datasets, MWM surpasses strong MIM baselines such as SparK, improving classification, detection, and segmentation and establishing text-driven generative pretraining as an effective strategy for medical imaging.

2 RELATED WORK

2.1 MASKED IMAGE MODELING

Masked modeling was first established in natural language processing by bidirectional encoder representations from transformers (BERT) [Devlin et al., (2019)], which learns contextual representations by masking tokens and predicting them during pretraining. [He et al., (2022)] transferred this principle to computer vision through MIM, where image regions are masked and reconstructed from the remaining visual context. This paradigm has become a strong generalization engine for both natural and medical images [Gupta et al., (2025); Liu et al., (2021); Huang et al., (2023)]. Representative methods include MAE [He et al., (2022)], MedMAE [Gupta et al., (2025); Xiao et al., (2023)], and SparK [Tian et al., (2023)]. MAE uses high-ratio random masking to push ViTs toward high-level semantic representations; MedMAE adapts this strategy to medical images; and SparK brings MIM to convolutional backbones by exploiting their multi-scale features.

Despite these advances, conventional MIM remains limited by random or structure-agnostic masking that ignores the semantic requirements of downstream tasks. Non-selective masking can force the model to spend pretraining capacity on irrelevant regions and thereby reduce transfer effectiveness. Recent region-aware masking methods recognize this limitation and attempt to improve semantic alignment along two main directions:

The first direction uses loss-driven adaptive masking. Methods such as hard patches mining (HPM) [Wang et al., (2023a)] and AnatoMask [Li et al., (2024)] dynamically analyze regional reconstruction errors and assign higher masking probabilities to high-loss areas, with the goal of emphasizing semantically valuable structures. This criterion, however, is not equivalent to task relevance:

it can redirect the model toward hard-to-reconstruct but irrelevant regions. In brain magnetic resonance imaging (MRI), for example, complex structures such as the spinal cord or eyeballs may attract unnecessary attention even when the downstream task is tumor detection.

The second direction incorporates external perceptual modules to guide masking. FocusMAE [Basu et al. (2024)], for instance, uses pretrained Region Proposal Networks (RPNs) to identify high-information regions for selective masking. This design can work well in specific settings, but its reliance on supervised RPN training with domain annotations restricts cross-task generalization.

These limitations make the central requirement clear: MIM needs a masking mechanism that is flexible, semantically aligned, and generalizable without task-specific annotations. Natural language provides precisely this kind of guidance because it is compact, interpretable, and directly expresses the semantic target of interest.

2.2 VISION-LANGUAGE PRE-TRAINED MODELS

Vision-language alignment models now make it practical to identify task-relevant visual regions under zero-shot and open-world settings [Radford et al. (2021)]. The Contrastive Language-Image Pretraining (CLIP) family learns a shared image-text embedding space through large-scale joint pretraining and has become a standard tool for zero-shot and open-set visual recognition. In medical imaging, BiomedCLIP [Zhang et al. (2023)] strengthens this cross-modal capability for biomedical concepts and delivers robust performance across downstream applications. Yet the pretraining value of these models remains underused. Existing studies mostly apply vision-language pretrained models to supervised downstream tasks such as object detection and semantic segmentation [Zhong et al. 2022, Aleem et al. 2024, Koleilat et al. 2024], leaving their open-vocabulary and zero-shot localization capabilities largely unexplored for masked image modeling.

In summary, current region-aware MIM methods improve over purely random masking but still fall short because they depend on additional supervision, confuse reconstruction difficulty with task relevance, or lack flexible semantic control. At the same time, vision-language models already provide the missing ingredient: prompt-conditioned semantic localization. MWM closes this gap by bringing natural language guidance into the MIM pretraining phase, enabling efficient and generalizable task-aware masking.

3 METHODOLOGY

3.1 OVERALL FRAMEWORK

As illustrated in Figure 1, **Mask What Matters (MWM)** is built around two core stages: (1) text-guided region localization and (2) region-aware masked image modeling. The first stage uses a frozen vision-language model to identify task-relevant regions from an open-vocabulary prompt. The second stage converts these regions into differentiated masking ratios, making the reconstruction objective explicitly focus on semantically important image content.

3.2 TEXT-GUIDED REGION LOCALIZATION

MWM localizes task-relevant regions through a deliberately modular pipeline. It begins with LLM-based prompt generation and uses BiomedCLIP to extract cross-modal embeddings. The method then generates a saliency map with the Multi-Modal Information Bottleneck (M2IB), binarizes it with K-Means, and refines it using the Segment Anything Model (SAM). Finally, MWM converts the refined mask into a bounding box with a controllable expansion margin, producing a robust spatial guide for masked modeling.

Prompt Generation. Textual prompts define the semantic target that MWM will localize, so prompt design is a key control interface rather than a peripheral detail. To provide rich semantic guidance, large language models (LLMs) are queried with structured prompts such as “Describe the typical visual characteristics of a [category] in a [modality] image”. Simple category phrases are also evaluated as compact alternatives. The experiments directly compare these prompt styles and quantify their effect on localization and downstream transfer.

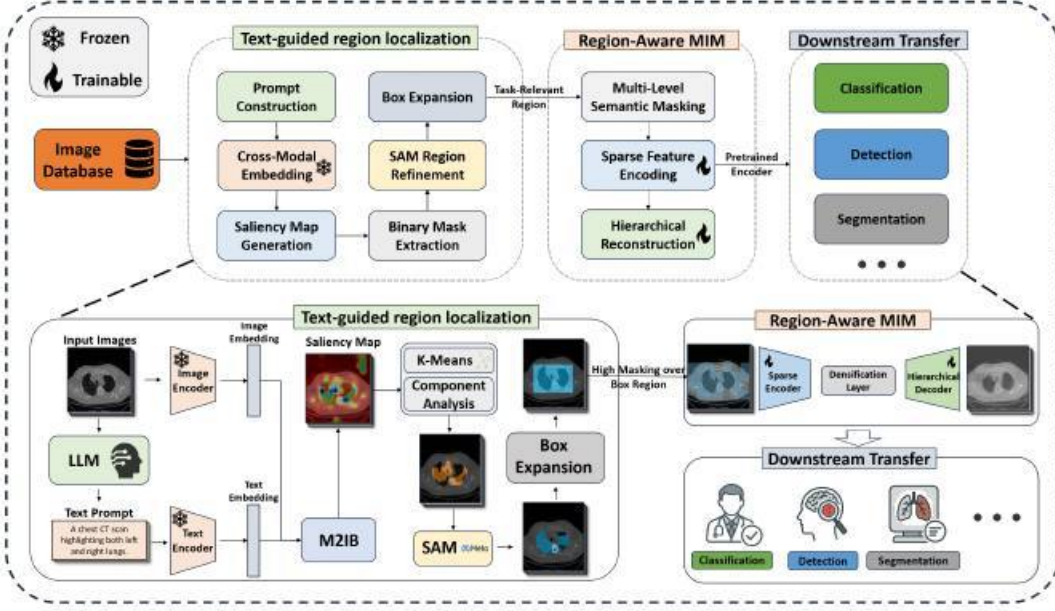


Figure 1: Overview of the MWM framework. The top panel presents the full three-stage pipeline: (1) text-guided region localization, (2) region-aware masked image modeling, and (3) downstream transfer. The bottom panel highlights the central mechanism: prompts identify semantically important regions that directly control the masking pattern.

Cross-Modal Embedding. After prompt generation, BiomedCLIP [Zhang et al., (2023)] encodes the medical image and prompt into a shared semantic space. Given an image I and a prompt T , the embeddings are computed as:

$$Z_{\text{img}} = \Phi_{\text{img}}(I), \quad Z_{\text{text}} = \Phi_{\text{text}}(T) \quad (1)$$

where Φ_{img} and Φ_{text} denote the vision and language encoders, respectively. These cross-modal embeddings anchor saliency estimation to the requested medical concept.

Saliency Map Generation. MWM uses a cross-modal saliency estimation module based on the Multi-modal Information Bottleneck (M2IB) [Wang et al., (2023b)] to identify visual regions that match the prompt while suppressing modality-specific redundancy. Formally, the saliency map $\lambda_S \in [0, 1]^{H \times W}$ assigns each spatial location an importance weight according to its semantic relevance to the prompt. M2IB optimizes the following objective:

$$\mathcal{L}_{\text{M2IB}} = \text{MI}(Z_{\text{img}}, Z_{\text{text}}) - \gamma \cdot \text{MI}(Z_{\text{img}}, I) \quad (2)$$

where Z_{img} and Z_{text} are image and text embeddings encoded by BiomedCLIP respectively, and γ controls the trade-off between preserving cross-modal relevance and filtering task-irrelevant visual information.

Binary Mask Extraction. MWM binarizes the saliency map with unsupervised K-Means clustering [Lloyd (1982)] to obtain a preliminary foreground mask for the prompt-guided region. The map is clustered into two pixel groups, and the higher-saliency cluster is selected as foreground. Connected component analysis then suppresses false activations by retaining the largest foreground regions, filtering noisy edges and concentrating the mask on the core semantic structures.

Region Refinement. MWM incorporates SAM as a refinement module to sharpen spatial accuracy [Kirillov et al., (2023)]. For each selected connected region $c_i \in \mathcal{C}^*$, its minimal enclosing bounding box is computed to serve as the visual prompt for SAM.

Given the original image I and the set of bounding boxes, SAM predicts a refined segmentation mask:

$$M_{\text{SAM}} = \text{SAM}(I, \text{Box}(c_i)) \quad (3)$$

The resulting mask M_{SAM} provides a more faithful spatial structure for region-aware masking.

Box-Based Region Expansion. The SAM mask can still contain boundary noise that distracts representation learning. To make region guidance robust, MWM converts the refined SAM mask into a bounding box with a controllable expansion margin, which serves as the final region of interest (ROI) for downstream masked modeling.

3.3 REGION-AWARE MASKED IMAGE MODELING

Given the region of interest generated by the localization module in Section 3.2, MWM performs masked image modeling with a three-stage pipeline: (1) multi-level masking with differentiated ratios based on region importance, (2) sparse encoding of only unmasked patches, and (3) hierarchical decoding that reconstructs the full image from sparse multi-scale features. This pipeline turns semantic localization into a concrete pretraining signal.

Multi-Level Semantic Masking. MWM uses multi-level semantic masking to operationalize the regions identified by text-driven localization. Patches highlighted by textual prompts, such as tumor- or organ-related regions, are masked at a higher ratio so the model must reconstruct task-relevant semantic features. Background regions are masked lightly or left unmasked, reducing redundant computation and preserving useful context. Figure 2 visualizes how this strategy differs from conventional random masking.

Sparse Feature Encoding. The visible patches form a sparse image that is encoded following the SparK design Tian et al. (2023):

$$\begin{aligned} X_{\text{sparse}} &= \{x_i \mid m_i = 1\}, \\ F &= \Phi_{\text{sparse}}(X_{\text{sparse}}) \end{aligned} \quad (4)$$

where x_i is the i -th patch, $m_i \in \{0, 1\}$ indicates visibility, Φ_{sparse} is the encoder, and $F = \{F_1, F_2, F_3, F_4\}$ are multi-scale features. Although our implementation instantiates MWM with convolutional backbones, the same interface extends to ViTs through token indexing.

Hierarchical Reconstruction. A lightweight UNet-style decoder reconstructs the full image through a top-down pathway. At each scale l , masked locations are filled with a learned embedding M_l :

$$F'_l(i) = \begin{cases} F_l(i), & \text{if patch } i \text{ is visible,} \\ M_l, & \text{if patch } i \text{ is masked.} \end{cases} \quad (5)$$

The densified map F'_l is projected by a scale-specific layer ϕ_l to obtain D_l , then upsampled and fused with features at the next lower level:

$$D_{l-1} = B_{l-1}(D_l) + \phi_{l-1}(F'_{l-1}). \quad (6)$$

The model is trained with mean-squared error on masked patches:

$$\mathcal{L}_{\text{recon}} = \frac{1}{|\mathcal{I}_{\text{mask}}|} \sum_{i \in \mathcal{I}_{\text{mask}}} \|\hat{x}_i - x_i\|^2, \quad (7)$$

where $\mathcal{I}_{\text{mask}}$ is the set of masked patch indices, x_i is the original patch, and $\hat{x}_i$ is its reconstruction.

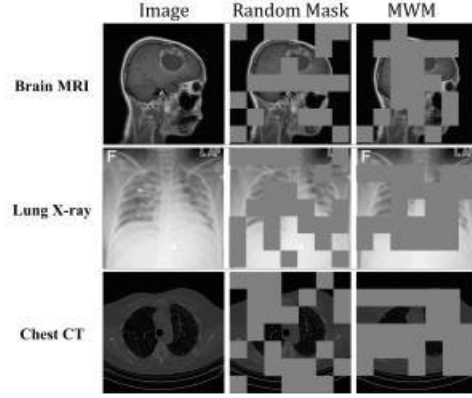


Figure 2: Comparison of masking strategies. Gray blocks mark the masked regions, highlighting how MWM concentrates masking on prompt-relevant content.

4 EXPERIMENTS

4.1 DATASETS

Self-Supervised Pretraining Datasets. Self-supervised pretraining uses three datasets that deliberately cover distinct medical imaging modalities. (1) *Brain MRI* contains approximately 17,000 images from Brain Tumor MRI Cheng et al. (2015), BRISC Fateh et al. (2025), and BraTS 2018 Menze

Method	Brain MRI		Chest CT		Lung X-ray	
	Prec.	Rec.	Prec.	Rec.	Prec.	Rec.
<i>Prompt Type</i>						
Phrase	0.22	0.86	0.21	0.99	0.42	0.82
Descrip. Sent.	0.30	0.94	0.37	0.88	0.44	0.66
<i>Impact of SAM</i>						
Coarse Ext.	0.22	0.95	0.36	0.87	0.37	0.84
+ SAM Ext.	0.30	0.94	0.37	0.88	0.42	0.82

Table 1: Text-guided localization performance (Occlusion Precision/Recall) across prompt types and refinement strategies. *Coarse Ext.* denotes bounding box extension of the coarse region; + *SAM Ext.* denotes additional SAM refinement.

et al. (2014), spanning diverse tumor types and anatomical structures. (2) *Lung X-ray* includes around 39,000 chest radiographs from COVID-QU-Ex Rahman et al. (2021); Chowdhury et al. (2020), covering COVID-19, non-COVID pneumonia, and normal cases. (3) *Chest CT* provides 12,000 slices from a lung disease dataset Konya (2020), with segmentation masks of fibrotic lungs from 107 patients.

Downstream Fine-Tuning Datasets. Downstream evaluation uses three datasets that test transfer across classification and detection-oriented settings. (1) *Brain Tumor MRI* Feltrin (2023) contains 4,479 MRI images across 44 tumor types, including astrocytoma, glioblastoma, meningioma, and ependymoma. (2) *Pediatric Lung X-ray* Kermay et al. (2018) contains 5,863 pediatric chest X-rays labeled as normal or pneumonia. (3) *Chest CT Cancer* Hany (2020) contains CT images from adenocarcinoma, large cell carcinoma, squamous cell carcinoma, and normal tissue classes.

4.2 EXPERIMENTAL SETUP

All self-supervised methods are evaluated under a controlled protocol with comparable backbone scale: convolution-based methods use ResNet-50 He et al. (2016), and transformer-based methods use ViT-S Dosovitskiy et al. (2020). The input resolution is 224×224, and all models are initialized with ImageNet-pretrained weights. Pretraining uses an initial learning rate of 2×10^{-4} . Downstream fine-tuning keeps the same encoder and input resolution. Experiments are run on NVIDIA Tesla A100 cards with 80 GB VRAM using Python 3.9, Ubuntu 22.04.3, and PyTorch 1.10.0. Unless otherwise specified, non-target variables such as prompt type and SAM usage are fixed to their empirically optimal settings, ensuring that comparisons isolate the factor under study.

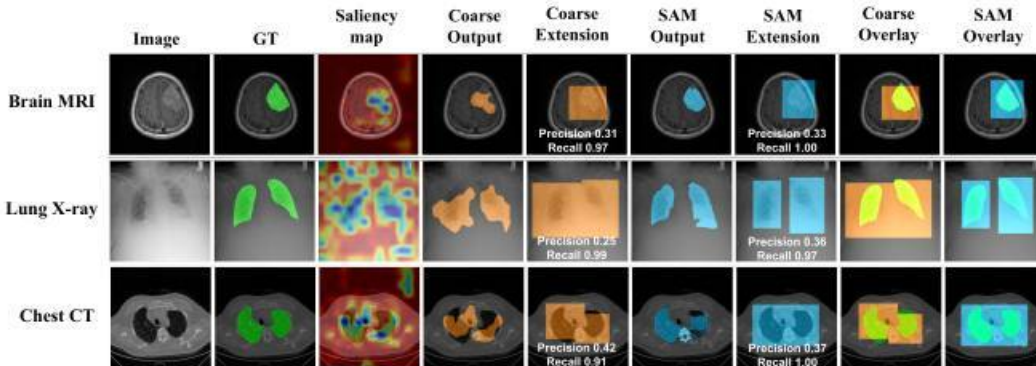


Figure 3: Representative text-guided localization results, showing that prompts recover task-relevant regions across modalities.

4.3 EVALUATION OF TEXT-GUIDED LOCALIZATION

We first evaluate whether the text-guided localization module reliably identifies task-relevant regions across brain MRI, lung X-ray, and chest CT. We also quantify how prompt type and SAM refinement affect localization. Predicted bounding boxes are compared with expert-annotated segmentation masks using occlusion precision ($|R_p \cap R_{gt}|/|R_p|$) and occlusion recall ($|R_p \cap R_{gt}|/|R_{gt}|$), where R_p is the predicted region and R_{gt} is the ground-truth annotation.

Table 1 shows that text-guided localization achieves consistently high occlusion recall across datasets (around 0.82) while retaining reasonable precision. This result confirms that MWM can identify task-relevant regions from textual descriptions alone, even under substantial modality-specific anatomical variation. Figure 3 provides representative visual examples.

Effect of Prompt Type on Region Localization. Prompt wording has a measurable and modality-dependent effect on localization. As shown in Table 1, descriptive sentence prompts perform better on Brain MRI and Chest CT, demonstrating that richer semantic descriptions improve localization when target structures are specific or spatially variable. In lung radiographs, concise phrases achieve higher recall (0.82 vs. 0.66), likely because the lungs occupy broad anatomical regions for which general terms are already informative.

Effect of SAM Refinement on Region Localization. SAM refinement strengthens the spatial quality of coarse masks. Table 1 shows that SAM improves precision while preserving high recall; for example, Brain MRI precision increases from 0.22 to 0.30 while recall remains above 0.94. This establishes SAM as a useful refinement step for converting prompt-driven saliency into sharper localization outputs.

4.4 COMPARISON WITH PREVIOUS SSL METHODS

We next evaluate the representation quality and generalization of MWM against representative SSL methods from both reconstruction-based and contrastive learning paradigms. All methods are pre-trained and evaluated with consistent datasets and training settings, making the comparison controlled and directly attributable to the pretraining strategy.

Image Classification. We evaluate image classification on three downstream medical imaging datasets under two standard settings: (1) *full fine-tuning*, where the entire model is updated; and (2) *linear probing*, where the encoder is frozen and only a linear classifier is trained. These two settings jointly test task adaptation and intrinsic representation quality.

- **Full fine-tuning:** Table 2 reports classification accuracy for MWM and representative self-supervised methods, including MAE [He et al. (2022)], SparK [Tian et al. (2023)], AnatoMask [Li et al. (2024)], SimCLR [Chen et al. (2020a)], MoCoV2 [Chen et al. (2020b)], and BYOL [Grill et al. (2020)]. MWM achieves the highest performance on every modality, demonstrating strong and consistent generalization. The largest gain appears on Chest CT, where MWM reaches 97.5% accuracy, surpassing SparK by +3.1 and AnatoMask by +1.7. This improvement directly supports the value of text-guided masking: MWM emphasizes semantically critical but spatially sparse regions, such as lung lobes in CT scans, while avoiding redundant background masking.

Method	Type	Brain MRI	Chest CT	Lung X-ray
MoCoV2	CL	95.3	91.7	93.1
BYOL	CL	91.3	93.7	93.9
SimCLR	CL	94.5	93.1	94.1
MAE	MIM	95.4	91.7	94.9
AnatoMask	MIM	96.5	95.8	95.2
SparK	MIM	96.2	94.4	94.7
MWM(Ours)	MIM	96.8	97.5	96.0

Table 2: Fine-tuning classification accuracy (%) on downstream datasets. MIM denotes Masked Image Modeling, and CL denotes Contrastive Learning. MWM achieves the strongest accuracy across all three modalities.

Method	Brain MRI	Chest CT	Lung X-ray
AnatoMask	70.2	63.9	85.3
SparK	69.8	63.5	81.6
MWM(Ours)	71.3	67.0	88.5

Table 3: Linear probing accuracy (%) on downstream datasets with a frozen encoder, showing the stronger representation quality learned by MWM.

Method	Detection		Segmentation	
	AP^{box}	AP_{75}^{box}	AP^{mask}	AP_{75}^{mask}
AnatoMask	45.2	53.1	46.2	54.1
SparK	45.6	53.2	44.8	52.0
MWM (Ours)	46.9	53.5	45.9	54.1

Table 4: Detection (AP^{box} , AP_{75}^{box}) and instance segmentation (AP^{mask} , AP_{75}^{mask}) results (%) on the BR35H dataset. MWM delivers the strongest box AP and competitive mask AP.

- **Linear probing:** Linear probing provides a stricter test of representation quality because the encoder remains frozen. We compare MWM with two recent masked image modeling methods, SparK and AnatoMask. As shown in Table 3, MWM outperforms both across all datasets, confirming that text-driven pretraining learns more effective transferable features.

Object Detection and Instance Segmentation. To test transfer beyond classification, we evaluate brain tumor detection and instance segmentation on the BR35H dataset [Hamada (2025)]. Table 4 shows that **MWM** achieves the highest AP^{box} (46.9%) and the second-highest AP^{mask} (45.9%), surpassing SparK by +1.3 and +1.1 points, respectively. These gains show that MWM representations transfer effectively to both object-level localization and pixel-wise segmentation.

Overall, the comparison establishes that text-driven masking makes pretraining more informative and transferable. The same semantic guidance improves classification, detection, and segmentation, demonstrating that the benefits of MWM are not confined to a single task type.

4.5 ABLATION STUDY

We conduct ablations to isolate the individual contributions of prompt design, SAM-based region refinement, and masking ratio to MWM’s downstream classification performance.

4.5.1 EFFECT OF PROMPT DESIGN AND REGION REFINEMENT

As shown in Table 5, descriptive sentence prompts outperform short phrases on brain MRI and chest CT, while lung X-ray favors the phrase prompt. This pattern matches the localization results in Section 4.3: prompts that produce more accurate region identification also produce stronger downstream performance. Crucially, both prompt types outperform random masking across all datasets. This result confirms that text-guided masking provides effective semantic alignment between pretraining and downstream tasks and that the benefit does not depend on a single prompt granularity. The SAM ablation further shows that adding this refinement module during pretraining consistently improves downstream classification, confirming that sharper spatial guidance strengthens text-guided masking.

4.5.2 EFFECT OF MASKING RATIO

As shown in Figure 4, MWM reaches 96.8% classification accuracy with only 40% masking, far below the roughly 70% ratio commonly used in masked image modeling. Even with this lower ratio, MWM exceeds SparK’s best result at 70%

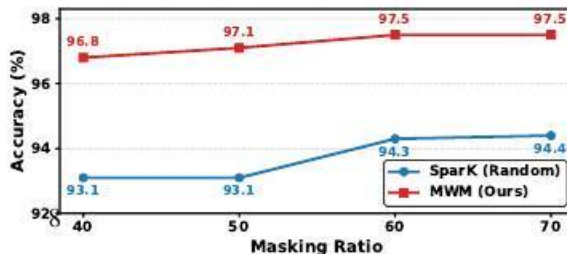


Figure 4: Classification performance on the Chest CT

Group	Setting	Brain MRI	Chest CT	Lung X-ray
Prompt Type	No Prompt	96.2	94.4	94.7
	Phrase	96.4	96.5	96.0
	Sentence	96.8	97.5	95.2
Impact of SAM	w/o SAM	96.4	96.8	95.8
	w/ SAM	96.8	97.5	96.0

Table 5: Downstream classification accuracy (%) under different prompt types and region refinement strategies across three imaging modalities. *No Prompt* refers to random masking without text-guided localization. *w/* and *w/o* denote with and without SAM, respectively; text guidance consistently improves over the no-prompt setting.

masking by +2.4% absolute accuracy. To ensure that this comparison is not caused by a poor SparK configuration, we evaluate SparK at 40%, 50%, 60%, and 70% masking ratios. The results confirm that 70% masking is the strongest SparK setting, supporting the robustness of the comparison. These results directly challenge the assumption that higher masking ratios are inherently better for MIM. MWM shows that semantically guided, task-aware masking can achieve stronger downstream performance with less masking by preserving task-relevant context and avoiding redundant reconstruction.

5 CONCLUSION AND LIMITATIONS

This work presents **Mask What Matters (MWM)**, a self-supervised pretraining framework that integrates text-guided semantic localization with region-aware masking. By using natural language prompts to identify task-relevant regions and applying differentiated masking, MWM turns semantic control into an explicit pretraining mechanism while requiring no per-image annotations.

Comprehensive experiments establish the effectiveness and generalizability of MWM: (1) natural language prompts reliably guide semantic region localization, validating text-driven masking as a practical mechanism; (2) MWM consistently outperforms existing methods such as SparK and AnatoMask in classification across three imaging modalities, while also improving detection and instance segmentation; (3) MWM maintains strong performance at lower masking ratios such as 40%, demonstrating that semantic guidance is more valuable than indiscriminately increasing the amount of masking.

Limitations. MWM also has clear limitations. First, although text prompts provide flexible control, this study does not systematically test robustness to prompt style, vocabulary variation, or noisy wording. Second, the framework depends on external textual descriptions, which may reduce reliability when suitable prompts are unavailable or inconsistent.

Looking forward, text-driven, region-aware masking offers a principled path toward more semantically grounded self-supervised learning. Extending MWM beyond medical imaging is a clear next step for broader vision–language pretraining.

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A USE OF LARGE LANGUAGE MODELS (LLMs)

LLMs were used *only* for language editing and presentation polishing, including grammar, phrasing, and minor stylistic clarity, of text written by the authors. They were *not* used for idea generation, experimental design, data annotation, code implementation, or analysis/interpretation of results. All technical content, algorithms, proofs, figures, tables, metrics, and conclusions were authored and verified by the authors. No confidential or identifying data were provided to LLMs beyond anonymized manuscript text.